Grade 8 Name_____________________
Unit 5 Date_____________________
Lesson 1 Practice Period_____________________

Use the table of possible solutions for questions 1–2.

Possible Solutions				
-11.5	36	$6\frac{3}{8}$	-8	-0.1212
-25	-9.5	-7.75	$6\frac{2}{7}$	-17.3

Determine the values from the table that are solutions to the inequalities.

1. $-3 + \frac{x}{2} < -7$

2. $-3 + \frac{x}{2} > -7$

3. Rewrite $-3 + \frac{x}{2} < -7$ and $-3 + \frac{x}{2} > -7$ so that -8 is a solution. Explain your reasoning.

4. Inequalities have ____________________ solutions that are represented on the number line.

5. Tanner and and Jonah were solving the inequality $-3 - \frac{y}{7} < -5$.
Jonah said the solution is $y < -14$, while Tanner said the solution is $y > -14$. Who do you agree with or do you agree with neither? Justify your reasoning mathematically.

6. When multiplying or dividing by negative coefficient in an inequality, the resulting inequality symbol ______________________________.

For questions 7–10, solve each inequality and graph the solution set.

7. $-8 \leq \frac{a}{-3} - 9$

8. $9m + 16 > -20$

9. $2.8 < -1.3p - 0.5$

10. $-2 + \frac{5}{4}x \geq -12$